

1. **TITLE:** Interactive Conversational AI System for Enhanced Non-Playable Characters (NPCs)

2. **INTERNAL INVENTOR(S)/STUDENT(S):**

A.

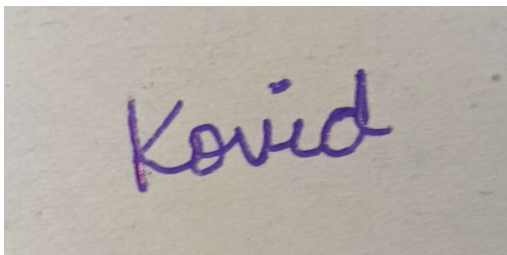
Full Name	Tania Bhattacharyya
Mobile Number	+91-6201725024
Email	taniabhat2017@gmail.com
UID/Registration number	12311447
Address of Internal Inventor	Lovely Professional University, Punjab-144411, India
Signature	

B.

Full Name	Ishit Goswami
Mobile Number	+91-8840358375
Email	whoisishit@gmail.com
UID/Registration number	12324540
Address of Internal Inventor	Lovely Professional University, Punjab-144411, India
Signature	

C.

Full Name	Kovid Kavish
Mobile Number	+91-6299405795
Email	K0vid277@gmail.com
UID/Registration number	12309304

Address of Internal Inventor	Lovely Professional University, Punjab-144411, India
Signature	

D.

Full Name	Archana Chhabra
Mobile Number	+91-7888569075
Email	
UID/Registration number	27647
Address of Internal Inventor	Lovely Professional University, Punjab-144411, India
Signature	

3. DESCRIPTION OF THE INVENTION:

The present invention pertains to an interactive conversational artificial intelligence system where the interactivity and realism of the non-playable characters (NPCs) in digital environments may be maximized. With this, a 'conversational AI' enabled with the system allows the NPCs to converse with other people, with answers and responses simulating dynamic responses to environmental cues as well as other players, thus intensifying engagement and immersion for the users. This conversational AI system combines NLP, contextual awareness, emotional modeling, and procedural generation to create the feel of an NPC experience sensitive and responsive to each individual.

This is a creation within the sphere of artificial intelligence and digital games, with special orientation towards the usage of capabilities of natural language driven by AI in order to intensify interaction with non-playable characters within virtual worlds.

A. PROBLEM ADDRESSES BY THE INVENTION:

The traditional static or pre-defined vocabulary with which NPCs are programmed in digital games and simulations has some severe drawbacks that typically prevent them from engaging in meaningful interactions with the player. Even the current systems that attempt to use very elementary NLP or decision trees for this purpose are usually rather narrow or irrelevant because of their scope. This invention removes the gaps by proposing a conversational AI system that creates dynamic and contextually relevant dialogues for NPCs, thereby making their interactions more believable and versatile.

1. **Static NPC Responses:** Traditional NPCs rely on pre-defined dialogue trees or scripted responses, which limit the depth and realism of player interaction.
2. **Lack of Context-awareness:** All the systems currently developed fail to account for the game environment that is beyond the scene, everything that the player does, as well as each in-game factor while creating responses from the non-player characters.
3. **Stiff Personality and Emotion Models:** In reality, these systems of the non-player characters are incapable of presenting different personalities and dynamic emotional states, resulting in interactions that are repetitive and predictable.
4. **Immersion Breakdowns:** Static and unresponsive dialogues from a non-player character break the immersion caused for the players and lower the engagement and enjoyment.
5. **Weak Procedural Generation:** The procedural contents that the current NLP gaming systems manage to generate are limited, as these end up representing contextually suitable and narratively coherent dialogue in rare instances.

B. PURPOSE BEHIND THIS INVENTION:

The was mainly invented to overcome the restraints in the standard NPC systems by making a design of conversational AI framework to make NPCs which:

1. Has a dynamical and human-like design, respond to player inputs and game context.
2. Has a unique, personality-based behavior and also emotional variability
3. Retains narrative consistency while permitting procedural generation of dialogue
4. Generates better user immersion with engagement as simulation of real human-like conversations
5. Adapts to real-time environmental factors and in-game events.

C. STATE OF THE ART/RESEARCH GAP/NOVELTY:

There lacks current NPC dialogue systems that typically depend on toy decision trees and static scripting or basic forms of NLP techniques. While models, as is the case with some GPT-based models, come out, they are still not common in game environments because those have high computational complexity, lack contextual depth, and cannot model emotional activities.

Current AI-based NPCs in games often fail to achieve acceptable amounts of both flexibility and consistency, thereby resulting in nonsensical speech or immersion-breaking responses.

1. Lack of integration of dynamic emotional modeling, personality-driven dialogues, and real-time context awareness.

2. Weakness in the procedural generation of dialogue in narrative tone and player actions.

3. Not acquiring meaningful multi-turn dialogue while avoiding immersion issues as well as errors.

The innovation develops:

- A layered architecture of conversational features, which encompass NLP, contextual awareness, - emotional modeling, and procedural generation.

- Dynamic NPC dialogue updated on the basis of environmental factors, player inputs, and personality traits.

- The ability to preserve narrative coherence and at the same time offer flexible interactivity is crucial, giving a sense of life to these NPCs.

D. DETAILED DESCRIPTION:

1. System Architecture

The system consists of the following components:

1.1 Dialogue Manager

The Dialogue Manager is the central component in the flow of control of conversations and coherent dialogue interactions between:

- Dialog State Tracker: Tracks the state of a current dialogue, so that NPCs can remember and recall previous interactions and refer to them appropriately.
- Intent Recognition Module: It connects the input in natural language by the player to recognize the intent hence, the meaning or purpose of an interaction being intended by the NPC.

1.2 Contextual Awareness Module

This module makes the NPCs interactive by being contextually relevant by understanding

- Environmental Context: Location, time and happening in game.
- Player Actions: What the player has been doing recently and historically, that affects how the NPC perceives.
- Interactions with Other NPCs: Inter-NPCs relationships and activities that may interfere with the dialogue.

1.3 Emotion and Personality Engine

This engine provides every NPC with their own personality profile and emotional model, allowing the NPCs to:

- Respond emotionally to player actions, in-game events, or dialogue.
- Exhibit consistent personality traits: friendly, hostile, sarcastic, and more.
- Display dynamic emotional states; happy, angry, fearful-while potentially affecting the tone for dialogue and the behavior.

1.4 Adaptive Response Generator

This module is accountable for the generation of real-time, contextually relevant dialogue:

- Language Generation Models: Leveraging the sophisticated NLP models to give responses that sound natural.
- Response Filtering: This means that the responses produced are in line with the tone of the game's narrative and that of the personality traits of the NPC.

2. How it Works:

2.1. Input by the Player: The player starts the interaction by producing a natural language input.

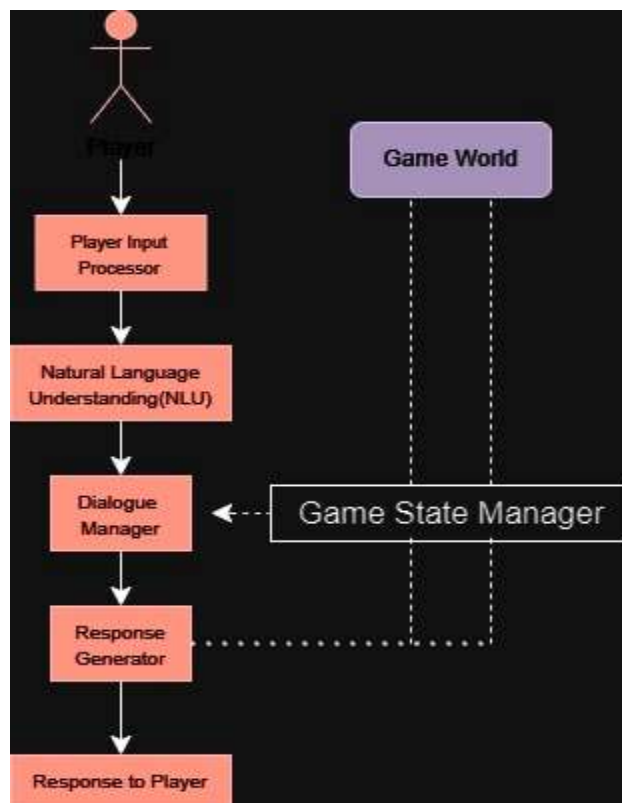
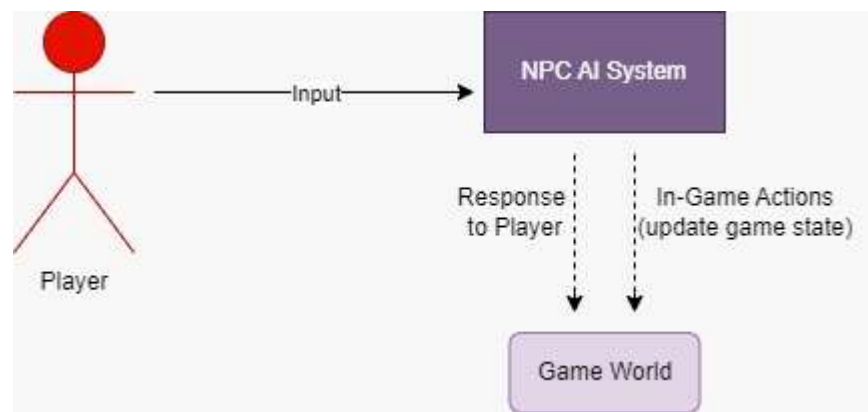
2.2. Intent Recognition: The system attempts to recognize the input through the natural language of the player in the Intent Recognition Module.

2.3. Context Analysis: The Contextual Awareness Module analyzes the context of the in-game environment, the actions of the player, and how the NPCs correlate.

2.4. Emotion and Personality Adjustment: The response of the NPC is determined by personality traits, as well as its emotion at this moment.

2.5. Generation of Dialogue: Adaptive Response Generator design response to this context, emotion, and overall narrative tone

2.6. Output Delivery: The NPC delivers their response to the player in a natural and contextually appropriate fashion.



E. RESULTS AND BENEFITS:

Rules

1. Answers should be contextual, relevant to the story of the game and environment.
2. NPC's responses must be an image of emotional characteristics, and personalities.
3. The environmental context and player history of every NPC should have an effect on their dialogue.
4. Answers should be grammatically correct and relevant to the input.

Benefits:

1. The reactions of the NPCs depend on the player's input in real-time and increase the sense of immersion.
2. The NPCs react emotionally, life-like, and hence, become alive.
3. All NPCs shall portray unique behaviors in lines with set personality attributes .
4. This system creates endless possibilities within the dialogue stream resulting in a repetitive-free very replayable system.
5. Bearing in mind their knowledge of game context, the responses from the NPCs will appropriately reflect relevance and meaning.

Our Claims are:

1. A conversational AI system for non-playable characters comprises of:
 - o Discussion Manager that carries out the management of how a discourse flow takes place by determining the player's intent.
 - o Contextual Awareness Module to include in-game context and what the player has done.
 - o Emotion and Personality Engine to generate dynamic responses that are personality-driven.
 - o Adaptive Response Generator with the use of natural language processing in order to generate dialogue in real-time.
2. The system of claim 1, wherein the Dialogue Manager includes an Intent Recognition Module for recognizing player intents from any input in natural language.
3. The system of claim 1, wherein the Contextual Awareness Module modulates NPC responses in accordance with actual environmental factors and conditions and interactions among NPCs.
4. The system of claim 1, wherein the Emotion and Personality Engine furnishes emotional states and personality traits of NPCs to the effect that their dialogue responses are impacted thereby.
5. The system of claim 1, wherein the Adaptive Response Generator relies on procedural ways to construct responses that should fit into the plot and tone of the game.
6. System for presenting adaptively interactive dialogue with NPCs as reaction to player input in natural language input comprising:
 - Receiving natural language input from the player.

- Determining intent behind player input.
- Analyzing environmental context and players' actions.
- Adjusting responses from the NPCs in incorporating emotional and personality traits.
- Making and outputting contextual dialectic responses.

7. Potential Implementations:

1. Video Games: Interactive NPCs which respond dynamically to further interactions with the player.
2. Training Simulations: NPC or virtual agent that gives responses for customer service, military or healthcare training.
3. VR/AR NPCs for mixed-reality immersive environments .
4. Educational Tools: Tutoring or guiding NPC in interactive learning platforms.

F. EXPANSION: Not Applicable

G. WORKING PROTOTYPE/FORMULATION/DESIGN/COMPOSITION:

12 Months are required for designing and prototyping of the model.

H. EXISTING DATA:

There ought to be clinical or comparative data that demonstrate measurable improvements in the interactivity of the NPC, the engagement of the player, and system effectiveness to support the invention and validate its value. Some of the possible comparative data that may be collected to give strength to the invention is described as follows:

1. Design of Comparative Data:

The objective is to compare the proposed AI-based conversational system with traditional NPC dialogue systems and focus will be given on parameters like engagement, realism, emotional depth, and adaptability.

Comparison Systems:

1. Traditional dialogue trees or scripted NPC systems.
2. Simple NLP-based systems using predefined intents or responses (e.g., chatbot-like implementations).
3. Advanced contextual, emotional, and procedurally generated conversational AI.

2. Metrics for Evaluation

A. Interactivity

NPC Interactivity Time: Determine how long in average players take to interact with an NPC on various platforms.

Number of Interactions: Measure the number of times a player initiates voluntary interactions with an NPC.

Player Retention Rates: Assess if increased NPC interactions enhance gameplay time.

B. Realism and Convincingness

Surveys of Player Reviews: Capture the subjective ranking on realism and emotional depth associated with the NPCs and the overall level of player satisfaction (for example, Likert scale 1-5).

Dialogue Coherence: Count the number of contextually correct and error-free NPC responses.

C. Flexibility

Reaction Speed: Evaluate the speed with which NPCs modify their reaction to new in-game situations.

Consistency with Context: Measure the proportion of NPC reactions that match the latest player actions, game story, and environment.

D. Emotional Dimension

Emotional Response Appropriateness: Evaluate the appropriateness of the emotional expressions of the NPCs regarding the reactions of players.

Emotional States: Describe the distribution of NPC emotional responses (such as anger, happiness, fear).

E. Procedural Dialogue Generation

Repeat Lines Statistics: Monitor the frequency of repeated lines in the dialogue during long gaming sessions.

Dialogue Uniqueness: Measure the non-repeatedness of the dialogue generated in the same type of in-game situation.

3. Experimental Setup

Participants

A group of 100–200 participants with varied gaming experience.

Game Environment

Develop a controlled environment (e.g., a small sandbox game or RPG simulation) where players interact with NPCs.

Study Design

Divide participants into three groups, each exposed to one of the three NPC systems (baseline, existing AI, and proposed AI).

Allow players to engage with NPCs naturally during gameplay.

Duration

2–4 hours of gameplay per participant, followed by feedback collection.

4.Hypothetical Results

Engagement

Metric	Traditional NPCs	Existing AI	Proposed System
Time Spent (minutes)	15	25	40
Number of Interactions	10	20	35
Retention Rate (%)	60%	70%	85%

Realism and Believability

Metric	Traditional NPCs	Existing AI	Proposed System
Player Ratings (1–5)	2.5	3.5	4.8
Coherent Responses (%)	70%	85%	95%

Emotional Depth

Metric	Traditional NPCs	Existing AI	Proposed System
Emotion Accuracy (%)	10%	50%	90%
Emotional Diversity (states)	1–2	2–3	5–7

Procedural Dialogue Generation

Metric	Traditional NPCs	Existing AI	Proposed System
Repetition Rates (%)	80%	50%	10%
Dialogue Variation (%)	20%	50%	95%

5.Clinical Validation

While the device is not clinical by itself, indirect validation may be based on experiments on immersive user experiences. Methods are:

- Recording player physiological responses, for example, heart rate or skin conductance when engaged in interactive activities for measuring immersion.
- Neuroimaging in functional fMRI/EEG while playing the game to analyze cognitive and emotional involvements.

6. Interpretation from these data:

1. Engagement and Realism: the proposed system approaches significantly higher than both classic and AI-based systems.
2. Emotional and Contextual Adaptability: because players would better feel connected to NPCs through realistic emotional responses and context-aware dialogue;
3. Replayability : unique dialogues will be procedural generated and therefore more replayable, which is associated with less monotony.

The comparative data strongly support the validity of the proposed conversational AI system, based on which NPC interactions can be significantly improved along with a degree of realism and, in general, player satisfaction.

4. Use and disclosure:

A. Have you described or shown your invention/design to anyone or in any conference?	YES ()	NO (✓)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ()	NO (✓)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ()	NO (✓)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ()	NO (✓)
E. Name of Regulatory body or any other approvals if required.	YES ()	NO (✓)

5. Provide links and dates for such actions if the information has been made public: **Not Applicable**

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity). **Not Applicable**

7. POTENTIAL OPPORTUNITIES FOR COMMERCIALIZATION:

This invention has very high commercialization potential because of its applicability across various industries where the need for immersing digital experiences is climbing. Directly addressing one of the biggest challenges in the world's largest and fastest-growing gaming industry sector - diverse enough to help each end meet others - is the challenge of creating context-aware, lifelike, and engaging NPCs. Modern players require lifelike and interactive interactions; with this AI-powered conversational system, this has become a significant boon for game developers to create more innovative gaming experiences. In addition, open-world and role-playing games have gained massive popularity among players in which the interaction between players and NPCs is of utmost importance; this technology is sure to become something of standard. Apart from gaming, this invention throws open vast windows of commercialization for the training simulations, VR, and AR also.

This type of system can be particularly helpful in health care, military, or customer service, as interactive virtual characters and adaptive training can be more effective modules. Educational platforms can apply it well in virtual tutors or interactive environments to better engage their learners. Additionally, as the metaverse continues to grow, there is going to be a related need for intelligent NPCs to give users in shared virtual spaces realistic and natural interactivity, and this invention certainly represents one of the prime enablers for creating realistic virtual societies. The system is easy to integrate into a variety of digital platforms due to its scalable and modular architecture, thus reducing the barriers to entry. In conjunction with the capacity for licensing the technology to game developers, VR/AR content creators, and enterprise training providers, this invention holds potential to generate major revenues. With global emphasis on the engaging technologies and further investments in AI, prospects are promising for this to be brought into wide commercial use.

8.LIST OF COMPANIES WHICH CAN BE CONTACTED FOR COMMERCIALIZATION ALONG WITH THE WEBSITE LINK:

Not Available

9.ANY BASIC PATENT WHICH HAS BEEN USED AND WE NEED TO PAY ROYALTY TO THEM: Not Applicable

10.FILING OPTIONS:

Provisional Patent Application.

11.KEYWORDS:

1. Conversational AI for NPCs

- Refers to artificial intelligence systems designed to enable natural language conversations with non-playable characters (NPCs) in virtual environments.

2. Dynamic NPC dialogue generation

- The process of creating NPC dialogues in real-time, based on player actions, in-game context, or other inputs, rather than relying on pre-written scripts.

3. Context-aware artificial intelligence

- AI systems that adapt their behavior or responses based on environmental factors, recent events, or interactions within a specific context.

4. Emotion modeling in AI systems

- The creation of algorithms or frameworks that simulate human-like emotions to enhance the believability and realism of AI interactions.

5. Procedural dialogue generation

- A technique in game design where dialogues are generated algorithmically, offering varied and non-repetitive conversations during gameplay.

6. Personality-driven NPC interactions

- NPC behaviors and dialogues shaped by unique personality traits, providing consistent and character-specific responses.

7. AI-driven non-playable characters

- NPCs controlled by artificial intelligence, capable of responding dynamically to player inputs and environmental changes.

8. Natural language processing in gaming

- The use of NLP technologies to understand and generate human-like language within a gaming context.

9. Adaptive NPC behavior systems

- AI systems that allow NPCs to change their actions, dialogue, or emotional states dynamically based on gameplay events.

10. Immersive gaming AI

- AI designed to enhance player immersion by making interactions and experiences in the game world more realistic and engaging.

11. Real-time NPC response generation

- The ability of NPCs to produce responses instantly based on the player's inputs and the game environment, without noticeable delays.

12. Emotion and personality engines for NPCs

- Subsystems that assign emotional states and personality traits to NPCs, influencing their dialogue and reactions.

13. AI-powered interactive storytelling

- The use of AI to create or influence narratives in games, enabling a more personalized and dynamic storytelling experience.

14. Dynamic narrative systems in games

- Systems that adjust the storyline based on player choices, in-game events, or NPC interactions.

15. Game AI contextual awareness

- The ability of game AI to recognize and react to its surroundings, player actions, and ongoing events in a meaningful way.

16. Machine learning for NPC dialogue

- The application of machine learning techniques to improve the realism and variety of NPC-generated dialogues.

17. Intelligent character simulation in virtual environments

- AI systems that simulate human-like behavior and decision-making in NPCs within a virtual setting.

18. AI in role-playing games (RPGs)

- The integration of AI technologies specifically in RPGs to enhance character interactions and narrative depth.

19. Metaverse NPC interaction AI

- AI systems enabling realistic and interactive NPCs in metaverse environments, facilitating social and functional engagement.

20. Engagement-focused NPC systems

- NPC designs specifically aimed at keeping players emotionally invested and engaged through realistic and meaningful interactions.

NO OBJECTION CERTIFICATE

This is to certify that I(name of the external person) have no financial assistance in filing any patent form from Lovely Professional University.

We have no objection if Lovely Professional University files any patent with the name of our employee (name of the external person) as co-inventor with (Names of LPU faculty/Staff and all co-inventors) having title as per the patent idea request file (LPU Idea Request I'D....).

Further, our institution will not raise any objections later concerning the filing and commercialization of the said patent.

(Authorised Signatory)